

Tutorial Sheet - 05

Section - A

Q.1 Solve $(D^2+6)y = \sin 4x$

$\Rightarrow D^2+6=0$

$D = \sqrt{-6}, D = \pm i\sqrt{6}$

Roots are complex and distinct

CF = $e^{ax} (C_1 \cos \sqrt{6}x + C_2 \sin \sqrt{6}x)$

CF = $C_1 \cos \sqrt{6}x + C_2 \sin \sqrt{6}x$

PI = $\frac{1}{D^2+6} \sin 4x$ $D^2 = -(a)^2$

$\frac{-1}{-16+6} \sin 4x \Rightarrow \frac{-1}{10} \sin 4x$

General solⁿ: $y = CF + PI$

$\Rightarrow C_1 \cos \sqrt{6}x + C_2 \sin \sqrt{6}x - \frac{1}{10} \sin 4x$

Q.2 Write the general formula PI of $f(D)y = x.v$ where v is function of x .

$\Rightarrow PI = \frac{1}{f(D)} x.v = x \frac{1}{f(D)} v - \frac{f'(D)}{[f(D)]^2} v$

Q.3 Write the general formula PI of $f(D)y = e^{ax}.v$, where v is a fun of x .

$\Rightarrow PI = \frac{1}{f(D)} e^{ax}.v = e^{ax} \frac{1}{f(D+a)} v$

Q.4 Define homogenous differential equation of second order.

⇒ Second order Differential equation is homogeneous if

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + c y = 0$$
 (No independent term)

Q.5 Define variation of parameter method.

⇒ General equation $\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Q y = R$

Sol of equation is of the form $y = Au + Bv$

u, v be known solution, find A, B by

$$A = \int \frac{u R}{uv' - vu'} dx + C_1, \quad B = \int \frac{-v R}{uv' - vu'} dx + C_2$$

Section - B

Q.1 Solve $(D^2 - 1)y = x \sin 3x + \cos x$

⇒ $D^2 - 1 = 0$

$D = \pm 1$

CF = $C_1 e^x + C_2 e^{-x}$

P.S ⇒ $\frac{1}{(D^2 - 1)} (x \sin 3x) + \frac{1}{(D^2 - 1)} (\cos x)$

$\frac{1}{(D^2 - 1)} (xv)$

$f(D) \Rightarrow x \frac{1}{(D^2 - 1)} \sin 3x - \frac{2D}{(D^2 - 1)} \sin 3x$

⇒ $\frac{-x}{10} \sin 3x - \frac{2 \cdot 3 \cos 3x}{(-3^2 - 1)^2} \Rightarrow \frac{-x}{10} \sin 3x + \frac{3}{50} \cos 3x$

$\frac{1}{(D^2 - 1)} \cos x = \frac{-1}{2} \cos x$

General solution $\Rightarrow C_1 e^x + C_2 e^{-x} - \frac{x}{10} \sin 3x + \frac{3}{50} \cos 3x - \frac{1}{2} \cos x$

Q.2 Solve $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + y = 2 \log x$

$\Rightarrow$ Equation is homogenous, differentiable equation with variable co-efficient

Put $x = e^z \Rightarrow z = \log x$ and $x \frac{dy}{dx} = Dy$, $x^2 \frac{d^2 y}{dx^2} = D(D-1)y$

Put values in equation

$D(D-1)y - Dy + y = 2z$ or $(D^2 - D - D + 1)y = 2z$

A.E. = $m^2 - 2m + 1 = 0$

$m = 1, 1$

C.F. = $(C_1 + C_2 z) e^z$

C.F. = $(C_1 + C_2 \log x) e^{\log x}$

P.I. = $\frac{1}{D^2 - 2D + 1} 2z$

$D^2 - 2D + 1$

$2[1 + (D^2 - 2D)]^{-1} \cdot z$

$2[1 - (D^2 - 2D)] \cdot z$

$2[z - (0 - 2)]$

$\Rightarrow 2(z + 2) \Rightarrow 2 \log x + 2$

$y = C.F. + P.I.$

$\Rightarrow x(C_1 + C_2 \log x) + 2(\log x + 2)$

Q.3 Solve by variation of parameters method, $\frac{d^2 y}{dx^2} - y = \frac{2}{1 + e^x}$

$\Rightarrow D^2 y - y = \frac{2}{1 + e^x}$

$(D^2 - 1)y = 0$

$m = \pm 1$

$$C.F = C_1 e^x + C_2 e^{-x}$$

$$u = e^x, \quad R = \frac{2}{1+e^x}$$

$$v = e^{-x}$$

$$y = Au + Bv$$

$$A = \int \frac{uR}{uv' - vu'} dx + C_1 \Rightarrow \int \frac{e^x \times \frac{2}{1+e^x}}{-e^x \cdot e^{-x} - e^{-x} \cdot e^x} dx + C_1$$

$$\Rightarrow \int \frac{e^x \times \frac{2}{1+e^x}}{-2} dx + C_1 \Rightarrow \int \frac{e^x}{1+e^x} dx + C_1$$

$$t = 1+e^x, \quad dt = e^x dx$$

$$A = - \int \frac{dt}{t} + C_1 \Rightarrow -\log t + C$$

$$A = -\log(1+e^x)$$

$$B = \int \frac{-vR}{uv' - vu'} dx + C_2 \Rightarrow \int \frac{-e^{-x} \cdot \frac{2}{1+e^x}}{-2} dx + C$$

$$B = \int \frac{1}{e^x(1+e^x)} dx$$

$$t = 1+e^x \Rightarrow dt = e^x dx$$

$$\int \frac{1}{e^x \cdot t} \cdot \frac{dt}{e^x} = \int \frac{1}{t \cdot 2e^x} dx$$

$$\Rightarrow \log(1+e^x) - x$$

$$y = Au + Bv$$

$$\Rightarrow \log(1+e^x) \cdot (e^x - e^{-x})$$

$$\text{General Solution: } C_1 + e^x + C_2 e^{-x} + \log(1+e^x)(e^x - e^{-x})$$

Section - C

Q.2 Solve $x^2 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + 2y = e^x$

$\Rightarrow$ eqⁿ is homogeneous, differentiable equation with variable coefficient
Put $x = e^z$, $z = \log x$, $x \frac{dy}{dx} = D y$, $x^2 \frac{d^2 y}{dx^2} = D(D-1)y$

$$D(D-1)y + 4Dy + 2y = e^x$$

$$(D^2 - D + 4D + 2)y = e^x \Rightarrow (D^2 + 3D + 2)y = e^{e^z} \quad \{x = e^z, e^x = e^{e^z}\}$$

$$m^2 + 3m + 2 = 0 \Rightarrow m = -1, -2$$

$$C.F. \Rightarrow C_1 e^{-z} + C_2 e^{-2z}$$

$$e^{-z} = \frac{1}{x}, \quad e^{-2z} = \frac{1}{x^2} \Rightarrow y_c \Rightarrow \frac{C_1}{x} + \frac{C_2}{x^2}$$

$$P.I = \frac{1}{D-a} f(x) = e^{ax} \int e^{-ax} f(x) dx$$

$$\frac{1}{D+a} f(x) = e^{-ax} \int e^{ax} f(x) dx$$

$$\frac{1}{(D+1)(D+2)} e^{e^z}$$

$$\left[\frac{1}{D+1} - \frac{1}{D+2} \right] e^{e^z} \quad \frac{1}{D+1} (e^{e^z}) = e^{-z} \int e^z \cdot e^{e^z} dz$$

$$\int e^z e^{e^z} dz$$

$$t = e^z, \quad dt = e^z dz$$

$$\int e^t dt = e^t = e^{e^z}$$

$$\frac{1}{D+1} (e^{e^z}) = e^{-z} \cdot e^{e^z}$$

$$\frac{1}{D+2} (e^{-z} e^{e^z}) = e^{-2z} \int e^{2z} \cdot e^{-z} e^{e^z} dz$$

$$e^{-2z} \int e^z e^{e^z} dz$$

$$P.I = e^{-2z} \cdot e^{e^z}$$

Converting back to $x \Rightarrow e^z = x, e^{-2z} = \frac{1}{x^2}$

$$y_p = \frac{e^x}{x^2}$$

$$\text{General solution} = \frac{C_1}{x} + \frac{C_2}{x} + \frac{e^x}{x^2}$$

Q.2 Solve $x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = 2x^2$

$\Rightarrow$ Equation is homogeneous, differential equation with variable coefficient.
Put $x = e^z \Rightarrow z = \log x$ and $x \frac{dy}{dx} = Dy, x^2 \frac{d^2 y}{dx^2} = D(D-1)y$

Put values

$$D(D-1)y - 3Dy + 4y = 2x^2 \Rightarrow (D^2 - 4D + 4)y = 2x^2$$

$$\text{So, } m^2 - 4m + 4 = 0 \Rightarrow m = 2, 2$$

$$y_c = (C_1 + C_2 z) e^{2z}$$

$$P.I = (D-2)y = 2e^{2z} \Rightarrow x^2 \cdot \frac{e^{2z}}{2!}$$

$$\left\{ \begin{aligned} (D-a)^n y &= e^{az} \\ \Rightarrow \frac{z^n}{n!} e^{az} \end{aligned} \right\}$$

$$\frac{z^2}{2} \cdot 2e^{2z} = z^2 \cdot e^{2z}$$

$$\text{General solution} \Rightarrow y = (C_1 + C_2 z) e^{2z} + z^2 e^{2z}$$

$$z = \log x, e^{2z} = x^2$$

$$y = x^2 (C_1 + C_2 \log x) + x^2 (\log x)^2$$

$$y = x^2 (C_1 + C_2 \log x + (\log x)^2)$$

Q.3 Solve by variation of parameter method $\frac{d^2 y}{dx^2} + a^2 y = \sec ax$

$$\Rightarrow \frac{d^2 y}{dx^2} + a^2 y = \sec ax$$

$$(D^2 + a^2)y = \sec ax$$

$$A.E \Rightarrow m^2 + a^2 = 0 \Rightarrow m = \pm ia$$

$$C.F = C_1 \cos ax + C_2 \sin ax$$

$$P.I = -y_1 \int \frac{y_2 x}{w} dx + y_2 \int \frac{y_1 x}{w} dx$$

$$y_1 = \cos ax, y_1' = -a \sin ax, y_2 = \sin ax, y_2' = a \cos ax$$

$$w = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \Rightarrow \begin{vmatrix} \cos ax & \sin ax \\ -a \sin ax & a \cos ax \end{vmatrix} \Rightarrow a$$

$$P.I = -\cos ax \int \frac{\sin ax}{a} \sec ax dx - \sin ax \int \frac{\cos ax}{a} \sec ax dx$$

$$\Rightarrow \frac{\cos ax}{a} \int \tan ax dx - \frac{\sin ax}{a} \int 1 dx$$

$$\Rightarrow -\frac{\cos ax}{a} \frac{1}{a} \log \cos ax - \frac{\sin ax}{a} \cdot x$$

$$\Rightarrow -\frac{x}{a} \sin ax + \frac{1}{a^2} \cos ax \cdot \log \cos ax$$



Complete solution $y_c = y_{C.F} + y_{P.I}$

$$\Rightarrow C_1 \cos ax + C_2 \sin ax + \frac{x \sin ax}{a} + \frac{1}{a^2} \cos ax \log \cos ax$$